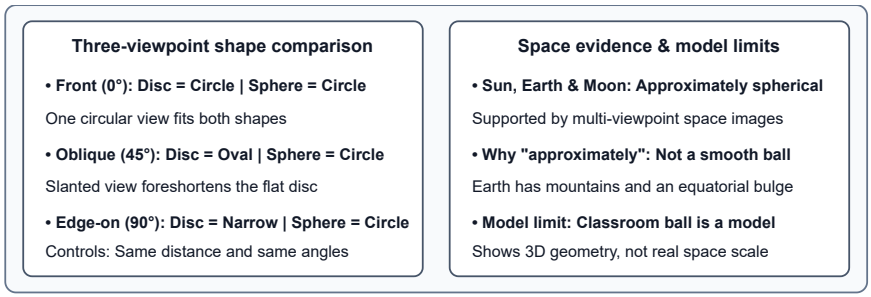
Reference sheets

Reference A · Three-viewpoint comparison: Flat disc vs Solid sphere

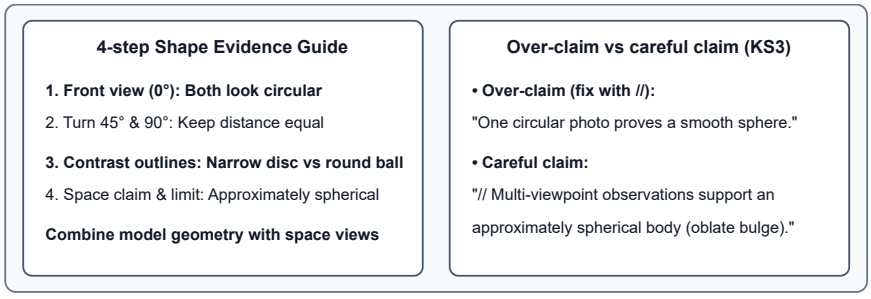


Three-viewpoint shape comparison. It compares a flat disc and a sphere at 0°, 45°, and 90°.

Reference only. Do not write on this sheet. Both objects are held at the same viewing distance across all three viewpoints.

Viewpoint angle.	Flat card disc outline.	Solid sphere (ball) outline.	What this viewpoint tells us.
1. Front view (0°).	Circular outline.	Circular outline.	Both shapes look circular from the front, so one view cannot separate them.
2. Oblique view (45°).	Oval (elliptical) outline.	Circular outline.	A slanted view foreshortens the flat disc while the sphere stays round.
3. Edge-on view (90° quarter-turn).	Narrow thin edge.	Circular outline.	Turning 90° separates the flat disc (narrow) from the 3D sphere (circular).

Reference B · Space evidence, careful claims, and model limits



Four-step Shape Evidence guide. It links 0°, 45°, and 90° views to spacecraft evidence and limits.

Evidence summary.	Key Stage 3 science meaning.	Honest model limit or careful wording.
E1 · Earth from spacecraft.	Spacecraft observations from multiple viewpoints show Earth	Earth has surface mountains and a small equatorial bulge,

Evidence summary.	Key Stage 3 science meaning.	Honest model limit or careful wording.
	as a three-dimensional round globe.	so it is approximately spherical.
E2 · Moon and Sun.	Multi-angle astronomical observations show the Moon and Sun have circular outlines from every viewpoint.	Described as approximately spherical three-dimensional bodies rather than flat discs.
E3 · Classroom ball model.	Predicts how a 3D sphere keeps a circular outline at 0°, 45°, and 90°.	Limit: the classroom ball is a geometric model, not real planetary proof or to scale.

Made by Matt

Tier pack · Aspire

Use Reference A and Reference B. Work in your book or on this sheet. Point, speak, sign, draw, type, write, or tell an adult. You may pass.

Arrival · 4 questions

1. Use Box 1 of the arrival drawing. Explain geometrically why a flat disc and a sphere share the same outline at 0°.

Help. Link the front cross-section to the line of sight.

2. Look at Box 2. Why does a 90° turn rule out a flat disc when an outline stays circular?

Help. Contrast 90° edge projection with spherical symmetry.

3. Look at Box 2 and Box 3. Name the two variables that must be controlled during a viewpoint comparison.

Help. State the two fairness controls.

4. Look at Box 3. Distinguish what a classroom ball test shows from real spacecraft evidence of Earth.

Help. Separate classroom geometry from real astronomical observations.

You do tasks

1. Use Reference A. Contrast the outlines of a flat disc and a sphere at 0°, 45°, and 90°. Keep viewing distance equal. **Staff I ☐ S ☐ C ☐**

Help. Link three-dimensional depth and viewpoint angle to the observed outlines.

2. Use Reference B. Evaluate the claim: 'One circular image proves a body is a perfect sphere.' Separate observation from interpretation. **Staff I ☐ S ☐ C ☐**

Help. Separate what the image shows from what it cannot prove.

3. Use Reference B. Link model evidence and spacecraft views to justify calling Sun, Earth, and Moon approximately spherical. State one model limit. **Staff I ☐ S ☐ C ☐**

Help. Link model predictions, multi-angle space telemetry, and oblate flattening.

Optional participation-loop prompt. Share your viewpoint comparison and limit with an adult. Hear one feedback step. Reply below or with // in your book. (Staff: the pupil responds to feedback with an adult, and R is marked only after the pupil has responded.)

Exit ticket

1. Why are multi-viewpoint spacecraft photos stronger evidence than one circular photo?

2. Explain why describing Earth as 'approximately spherical' is more accurate than 'a perfect sphere'.

3. State what the classroom disc-and-sphere comparison proves and what it cannot prove alone.

Pupil voice

What helped you today, and what would you change?

Point, speak, sign, draw, write or choose. You may pass.

Feedback and response · Aspire

Feedback codes

Staff I ☐ S ☐ C ☐ · Route card feedback boxes: VF ☐ · WS ☐ · I ☐ · E ☐ · NS + … ☐ · R ☐ · // ☐ · ? ☐

Code	Meaning
VF	Verbal feedback
WS	Worked with support
I	Independent
NS + …	Next step
E	Evidence captured
R	Pupil responded to feedback
//	Self-edit point
?	Read this back to me

My response

My response: Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Made by Matt